Optimization of Heterogeneous Edge AI Inference Pipeline on TDA4VM using C7x DSP

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***Abstract*—Deploying deep learning models on edge devices requires balancing high inference accuracy with strict power, thermal, and latency constraints. While standard edge CPUs or GPUs offer deployment flexibility, they often fail to meet the stringent power-efficiency requirements of advanced industrial vision applications. The Texas Instruments TDA4VM System-on-Chip (SoC) addresses this challenge through a heterogeneous architecture featuring a C7x DSP paired with a Matrix Multiply Accelerator (MMA). This project proposes an optimized heterogeneous inference pipeline tailored for the TDA4VM SoC in order to maximize throughput and energy efficiency, using the TIDL toolkit for INT8 post-training quantization.**

***Index Terms*—edge AI, TDA4VM, C7x DSP, heterogeneous inference, INT8 quantization, TIDL, MSMC, real-time processing.**

I. Introduction

Deploying deep learning models on edge devices requires balancing high inference accuracy with strict power, thermal, and latency constraints. While standard edge CPUs or GPUs offer deployment flexibility, they often fail to meet the stringent power-efficiency requirements of advanced industrial vision applications. The Texas Instruments TDA4VM System-on-Chip (SoC) addresses this challenge through a heterogeneous architecture featuring a C7x DSP paired with a Matrix Multiply Accelerator (MMA). However, fully utilizing this architecture presents several software engineering challenges, including cross-core synchronization, memory bandwidth bottlenecks between DDR and on-chip SRAM, and accuracy degradation caused by model quantization.

II. Related Work

Deploying deep neural networks on edge computing platforms has been extensively studied to address the trade-off between inference accuracy and hardware constraints. Wang et al. [1] provide a comprehensive survey on edge intelligence, highlighting that while standard CPUs and GPUs offer flexibility, they struggle with the strict power limitations of mobile and automotive systems, necessitating heterogeneous SoC designs.

To bridge the gap between heavy deep learning models and low-power edge processors, post-training quantization (PTQ) and integer-only inference have become industry standards. Jacob et al. [2] introduced a foundational quantization scheme that maps FP32 weights and activations to 8-bit integers, significantly reducing latency and memory bandwidth requirements without substantial accuracy drops. This methodology is directly applicable to the INT8 optimization pipeline executed on the C7x DSP and MMA.

In the context of specific vision tasks, such as Infrared Small Target Detection (IRSTD), achieving high precision often relies on complex feature extraction mechanisms. For instance, Li et al. [3] proposed DNA-Net, a dense nested attention network that achieves state-of-the-art accuracy in detecting dim and small targets. However, such intricate architectures are typically designed for high-end GPUs. This project builds upon these hardware and software foundations by taking computationally intensive, state-of-the-art vision models and mapping them onto the highly optimized, heterogeneous architecture of the TDA4VM.

III. Problem Definition

A. System / Application Description

B. Objectives

C. Constraints

D. Mathematical Formulation (Optimization Track)

IV. System Architecture and Design

A. Hardware/Software Architecture

The proposed system is designed as an end-to-end partially asynchronous data pipeline. The major components of the system are listed below:

- **Data Ingestion and Pre-processing Unit (ARM Cortex-A72):** Responsible for capturing input frames from the sensor interface and preparing the data layout required by the neural network (e.g., converting from NHWC to NCHW format).
- **TIDL Execution Engine (C7x DSP + MMA):** Executes the compiled INT8 neural network layers using the optimized inference kernels provided by the TIDL framework.
- **Memory Management Unit:** Utilizes the Multicore Shared Memory Controller (MSMC) to cache intermediate feature maps, thereby reducing costly accesses to external DDR memory. DDR-to-MSMC transfers are managed by the EDMA controller using asynchronous

pre-fetching. Intermediate tensors with high access frequency are pinned to MSMC SRAM, while low-reuse tensors remain in DDR to conserve the limited 8 MB on-chip capacity. This placement policy is determined empirically during the profiling phase.

- **Post-processing Unit (C66x DSP / ARM):** Processes the network outputs and converts them into actionable outputs such as bounding boxes, object labels, or detection metrics.

B. Dataflow and System Model

The core optimization strategy relies on workload distribution across heterogeneous processing units. Compute-intensive tensor operations will be offloaded to the C7x DSP together with the MMA accelerator, while sequential tasks such as image pre-processing and post-processing will be executed on the ARM Cortex-A72 or the C66x DSP cores.

Task partitioning decisions are made *statically* at compile time, based on the computational profile of each pipeline stage obtained during baseline profiling. Compute-bound layers such as convolutions and matrix multiplications are assigned to the C7x DSP and MMA, while memory-bound or control-flow-heavy tasks such as NMS decoding and format conversion are assigned to the ARM or C66x cores. This assignment is validated using TI SysBIOS profiling utilities; if runtime load imbalance is observed, an adaptive round-robin policy on the C66x cores will be explored.

The pipeline operates in a *partially asynchronous* manner. Inter-stage communication relies on *double-buffering* through shared MSMC SRAM regions, so that one buffer is written by the upstream stage while the downstream stage reads from the other simultaneously. Synchronization between cores is achieved via hardware semaphores provided by the TI SysBIOS IPC framework, which avoids costly busy-wait polling. Data movement between DDR and MSMC SRAM is handled by the EDMA controller: DMA transfers are issued asynchronously so that memory pre-fetching of the next input tile overlaps with the C7x DSP processing the current tile, minimising pipeline stalls due to memory latency.

C. Requirements Analysis (Implementation Track)

1) Functional Requirements:

2) Non-Functional Requirements:

D. Timing and Execution Model

E. Model of Computation

F. Assumptions and Design Limitations

G. Resource Model

This project proposes an optimized heterogeneous inference pipeline tailored for the TDA4VM SoC in order to maximize throughput and energy efficiency. The methodology focuses on migrating a vision-based deep learning model (e.g., a lightweight CNN for object detection) to the C7x DSP. The Texas Instruments Deep Learning (TIDL) toolkit will be used to perform Post-Training Quantization (PTQ), converting the baseline FP32 model to an INT8 format. Table I summarises the candidate models considered for deployment, comparing their estimated activation memory footprint against the 8 MB MSMC SRAM budget and their reported COCO mAP accuracy.

TABLE I
CANDIDATE MODEL QUICK-REFERENCE: ACTIVATION MEMORY VS. MSMC SRAM FIT AND ACCURACY.

Model	Input Size	Est. Act. Memory	SRAM Fit	mAP (COCO)
YOLOX-Nano	416×416	∼2–3 MB	Full	∼25
YOLOX-Tiny	416×416	∼3–4 MB	Full	∼32
YOLOX-S	640×640	∼6–8 MB	Marginal	∼40
SSD-MobileNetV2	300×300	∼3 MB	Full	∼22
EfficientDet-Lite0	320×320	∼3–4 MB	Full	∼26
EfficientDet-Lite1	384×384	∼5–6 MB	Likely	∼31
PeleeNet-SSD	304×304	∼3–4 MB	Full	∼22

V. METHODOLOGY

A. Implementation Details (Implementation Track)

B. Optimization Pipeline (Optimization Track)

- **Milestone 1 (Weeks 1–2):** Baseline model selection, FP32 training and validation, and development environment setup (Processor SDK and Code Composer Studio). Optimization objectives, decision variables, and evaluation metrics (latency, FPS, TOPS/W, mAP) will be formally defined at this stage.
- **Milestone 2 (Weeks 3–4):** Model export to ONNX format, TIDL model import, INT8 quantization, and accuracy evaluation using the PC host emulator.
- **Milestone 3 (Weeks 5–6):** Implementation of the heterogeneous inference pipeline on the target hardware and distribution of tasks across ARM, C66x, and C7x cores. MSMC double-buffering and EDMA-based asynchronous data transfer scheduling will be configured, and IPC semaphore synchronization will be implemented at this stage.
- **Milestone 4 (Weeks 7–8):** System profiling using performance counters, benchmarking (latency, FPS, and MAC utilization), and preparation of the final project report. Memory placement will be optimised based on profiling data; per-stage latency and bandwidth will be measured; and latency-energy trade-offs will be analysed by comparing the optimised system against the baseline.

C. Algorithm Description

VI. EXPERIMENTAL SETUP

A. Platform and Tools

B. Datasets / Workloads

C. Evaluation Metrics

VII. SYSTEM-LEVEL CONSIDERATIONS

Edge AI is vital for modern autonomous systems, but running accurate deep learning models on small edge devices is difficult due to strict power and heat limits. Solving this optimization problem on the TI TDA4VM chip is not just a

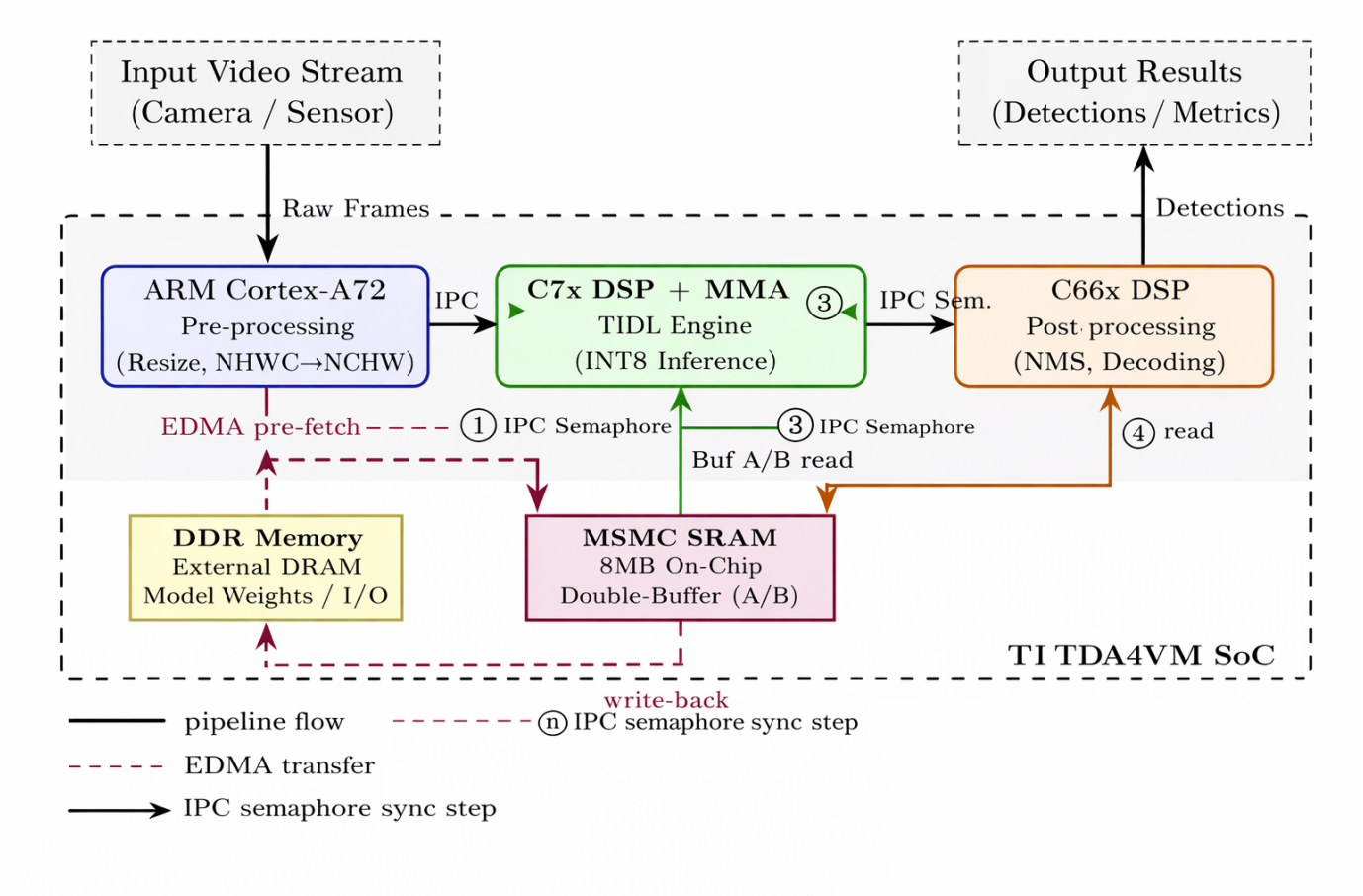

Fig. 1. Heterogeneous Edge AI inference pipeline on the TI TDA4VM SoC.

technical win; it directly improves public safety, saves energy, and lowers costs.

Safety and Reliability: In real-world applications like self-driving cars (ADAS) or rescue drones, fast and local data processing saves lives. For example, detecting a pedestrian in low light requires quick and accurate vision models. If the system is slow, it can cause fatal accidents. By optimizing the software to run efficiently on the C7x DSP, this project guarantees fast, real-time reactions, making autonomous vehicles much safer.

Sustainability and Cost Reduction: Traditional graphics cards (GPUs) use too much power, draining the batteries of electric cars and drones quickly. By shifting the heavy AI calculations to the highly efficient C7x DSP using INT8 quantization, we drastically cut down energy use. This efficiency also lowers hardware costs, allowing standard, affordable cars to have advanced safety features that are usually limited to luxury vehicles.

Beneficiaries and Impact Measurement: The main beneficiaries are everyday drivers, pedestrians, and emergency teams who rely on safe technology. We will measure this project's success using three clear metrics:

- **Safety:** Response time (Latency in milliseconds) and speed (Frames Per Second).
- **Sustainability:** Energy efficiency measured in performance per watt (TOPS/W).
- **Reliability:** The model's accuracy (e.g., mAP) after the optimization process.

The following design considerations are explicitly addressed as evaluation dimensions of this project. Memory bandwidth limitations will be measured using TI hardware performance counters to quantify the improvement achieved by MSMC placement over DDR. The latency-throughput trade-off will be analysed by varying pipeline buffer depth and reporting both single-frame latency and sustained FPS. Synchronization overhead between ARM, C7x, and C66x cores will be profiled to verify that asynchronous scheduling eliminates bottlenecks. The impact of INT8 quantization on accuracy relative to the FP32 baseline will be reported as an accuracy-latency curve. Energy efficiency in TOPS/W will be estimated from TDA4VM power monitoring registers. Finally, scalability will be evaluated by testing the pipeline at different input resolutions.

Acknowledgments

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